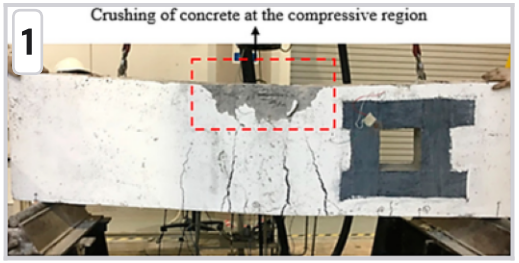
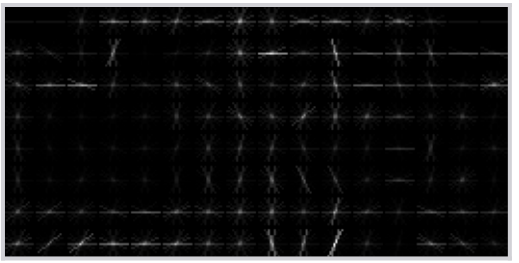
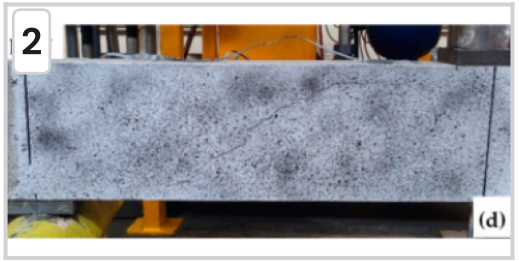
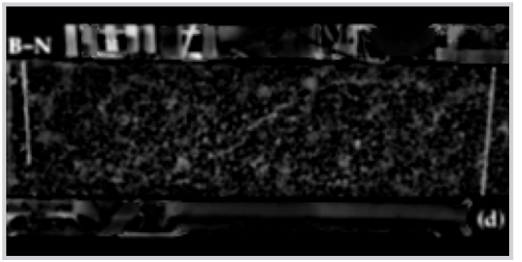
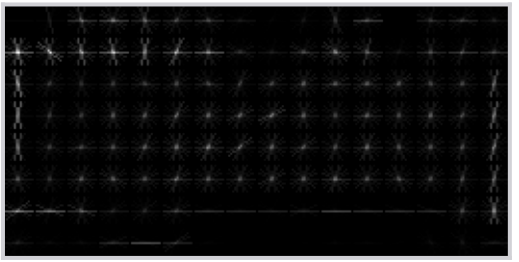
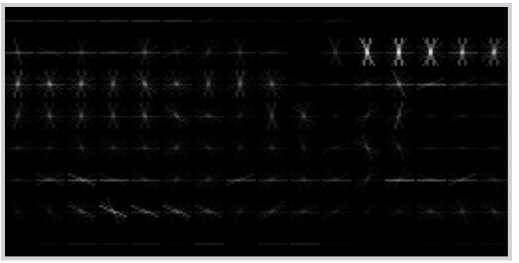
Preprocessing makes cracks bright lines, and HOG records their direction in every 16-pixel cell.

	Image	Preprocessed	HOG
Flexural			
Shear			
Shear-flexural			

Source: 295 images, 275 specimens, 50 open-access articles